

V4.2 K_V42 v1 Comparative Report

Shapiro-Calibrated Graph Curvature / Focusing Index

Purpose:
Correct K_V42 v0, which over-rewarded pure connectivity shortcuts.

New v1 formula:
 $K_{v1} = F_{in} + F_{bundle} - 2P_{core} - 2P_{boundary} - 5P_{shortcut} - P_{wormhole}$

Formula meanings:
F_in = inward movement of off-axis paths near the source.
F_bundle = convergence/narrowing of the path bundle behind the source.
P_core = fraction of paths entering $r \leq 2.0$; penalizes core collapse/wormhole behaviour.
P_boundary = fraction of paths touching $|y| \geq 9$; penalizes wall/boundary fleeing.
P_shortcut = excessive scout shortcut beyond allowed gain; now weighted by 5.
P_wormhole = $abs(avg_bundle_delay)*5$ when $avg_bundle_delay < 0$; requires net friction.

Guardrail:
K_v1 is still Ricci curvature, not GR curvature, and not gravitational lensing.
It is a stricter operational diagnostic.

Top cases by K_v1:

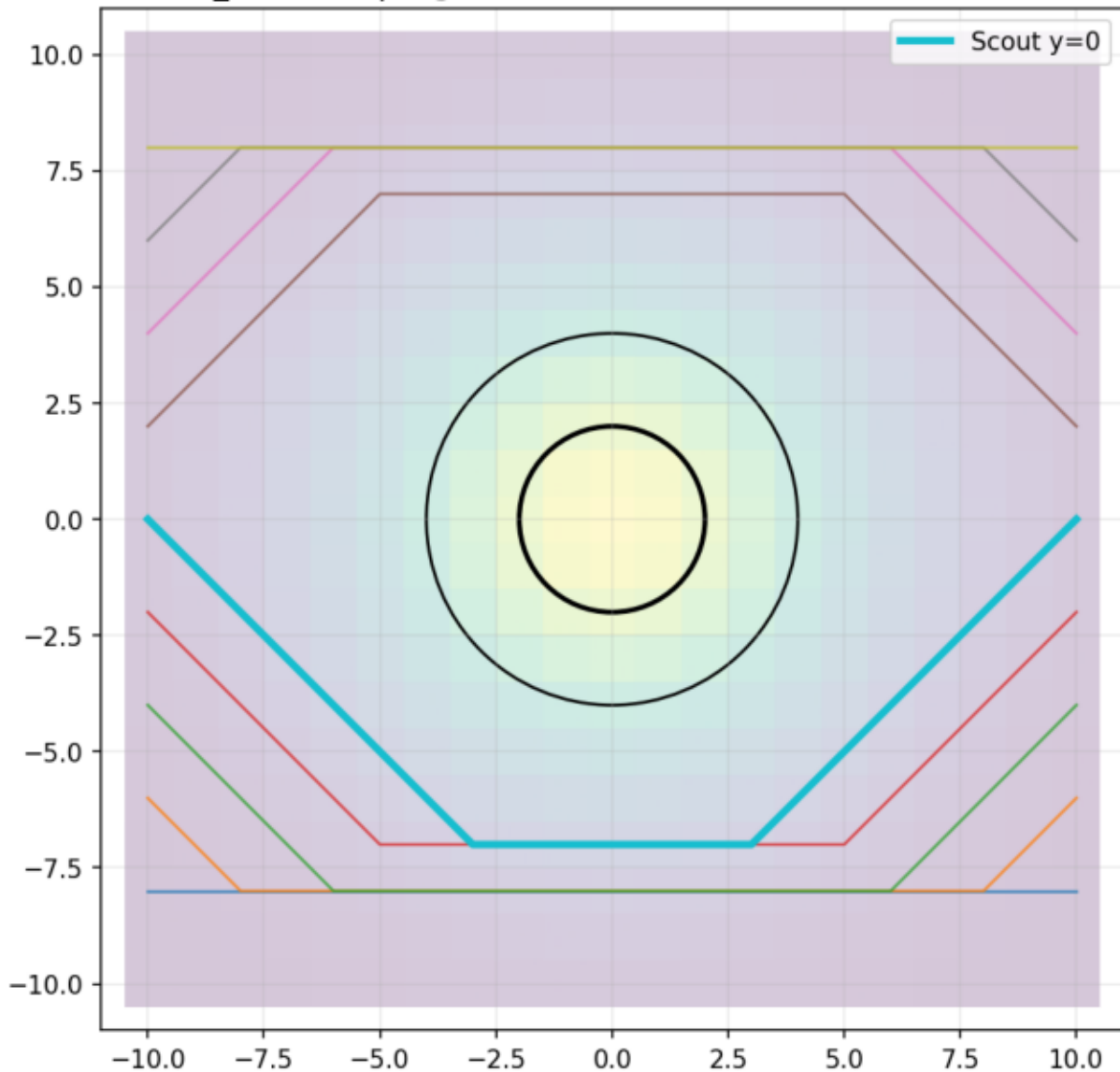
1D-C1_l1_m3: K=0.14, avgD=2.18, Fin=1.25, Fb=0.00, Pw=0.00
1D-A_lambda0: K=-0.67, avgD=0.00, Fin=0.00, Fb=0.00, Pw=0.00
1D-B_flat: K=-0.67, avgD=0.00, Fin=0.00, Fb=0.00, Pw=0.00
1D-B_equal_l1_m1: K=-0.67, avgD=0.00, Fin=0.00, Fb=0.00, Pw=0.00
1D-B_delay_l2_m1: K=-0.67, avgD=1.51, Fin=0.00, Fb=0.00, Pw=0.00
1D-C1_flat: K=-0.67, avgD=0.00, Fin=0.00, Fb=0.00, Pw=0.00
1D-C2_flat: K=-0.67, avgD=0.00, Fin=0.00, Fb=0.00, Pw=0.00
1D-C2_lambda0_only: K=-0.67, avgD=2.70, Fin=0.00, Fb=0.00, Pw=0.00
1D-C3_flat: K=-0.67, avgD=0.00, Fin=0.00, Fb=0.00, Pw=0.00
1D-C3_lambda0_only: K=-0.67, avgD=2.70, Fin=0.00, Fb=0.00, Pw=0.00

v0 false winner: 1D-C3_connect_mu2 now K_v1 = -35.50
Preserved candidate: 1D-C3_lamr2_mu3 now K_v1 = -18.16
Top v1 case: 1D-C1_l1_m3 with K_v1 = 0.14

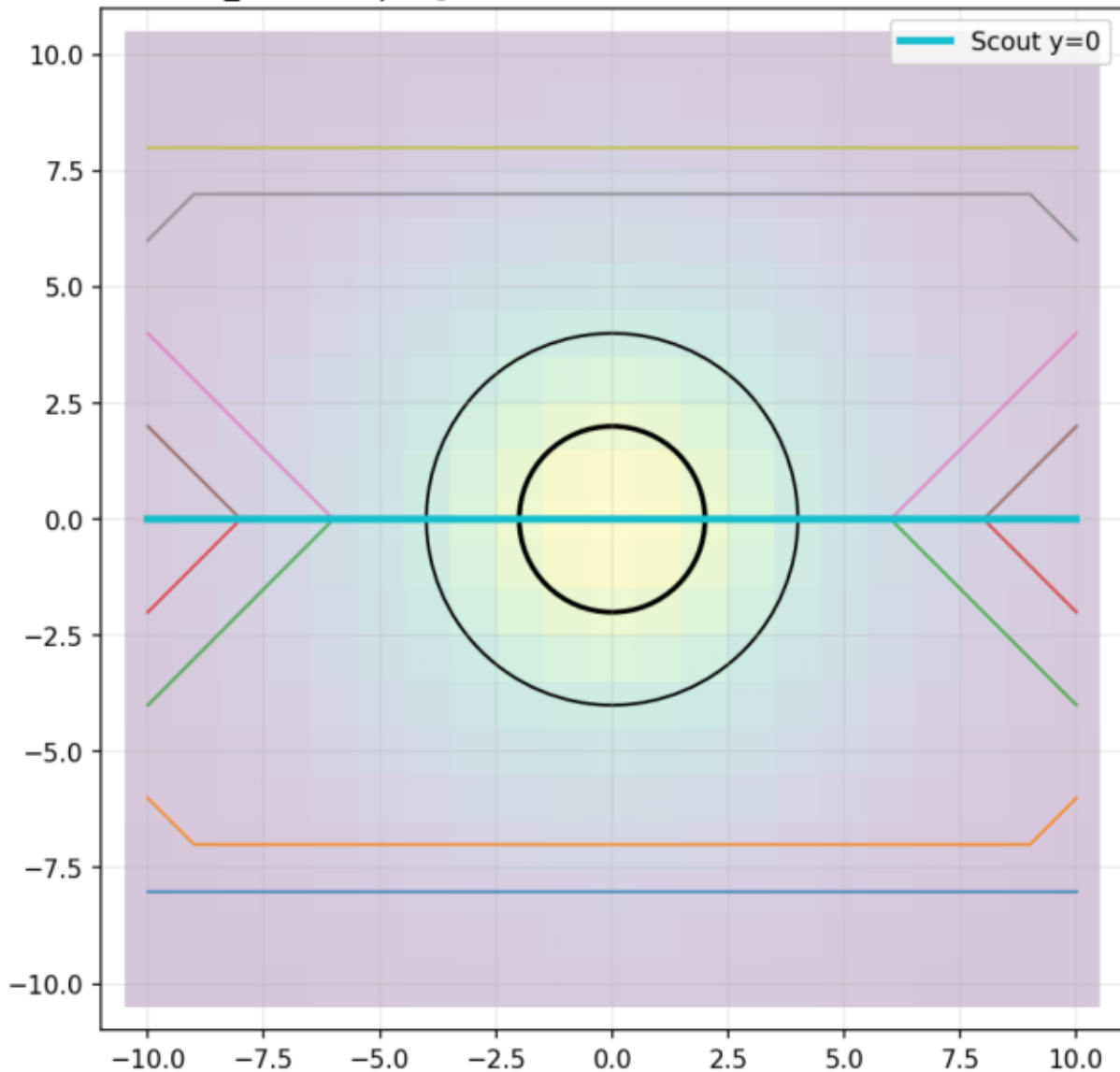
K_V42 v1 Ranking Table - Top 20

01.	1D-C1_l1_m3	K=	0.14	avgD=	2.18	Fin=	1.25	Fb=	0.00	Pw=	0.00
02.	1D-A_lambda0	K=	-0.67	avgD=	0.00	Fin=	0.00	Fb=	0.00	Pw=	0.00
03.	1D-B_flat	K=	-0.67	avgD=	0.00	Fin=	0.00	Fb=	0.00	Pw=	0.00
04.	1D-B_equal_l1_m1	K=	-0.67	avgD=	0.00	Fin=	0.00	Fb=	0.00	Pw=	0.00
05.	1D-B_delay_l2_m1	K=	-0.67	avgD=	1.51	Fin=	0.00	Fb=	0.00	Pw=	0.00
06.	1D-C1_flat	K=	-0.67	avgD=	0.00	Fin=	0.00	Fb=	0.00	Pw=	0.00
07.	1D-C2_flat	K=	-0.67	avgD=	0.00	Fin=	0.00	Fb=	0.00	Pw=	0.00
08.	1D-C2_lambda0_only	K=	-0.67	avgD=	2.70	Fin=	0.00	Fb=	0.00	Pw=	0.00
09.	1D-C3_flat	K=	-0.67	avgD=	0.00	Fin=	0.00	Fb=	0.00	Pw=	0.00
10.	1D-C3_lambda0_only	K=	-0.67	avgD=	2.70	Fin=	0.00	Fb=	0.00	Pw=	0.00
11.	1D-C3_core_delay_only	K=	-0.75	avgD=	3.07	Fin=-0.75	Fb=	0.00	Pw=	0.00	
12.	1D-A_lambda1	K=	-1.00	avgD=	1.76	Fin=-1.00	Fb=	0.00	Pw=	0.00	
13.	1D-C1_l1_m1	K=	-1.17	avgD=	3.84	Fin=-0.50	Fb=	0.00	Pw=	0.00	
14.	1D-C1_l1_m2	K=	-1.17	avgD=	2.97	Fin=-0.50	Fb=	0.00	Pw=	0.00	
15.	1D-C2_radial_delay_only	K=	-1.75	avgD=	3.73	Fin=-1.75	Fb=	0.00	Pw=	0.00	
16.	1D-C2_balanced_lamr1_mu1	K=	-1.75	avgD=	3.72	Fin=-1.75	Fb=	0.00	Pw=	0.00	
17.	1D-C2_focus_lamr1_mu2	K=	-1.97	avgD=	3.37	Fin=-1.75	Fb=	0.00	Pw=	0.00	
18.	1D-C1_delay_l1	K=	-2.00	avgD=	4.67	Fin=-2.00	Fb=	0.00	Pw=	0.00	
19.	1D-A_lambda5	K=	-2.25	avgD=	2.95	Fin=-2.25	Fb=	0.00	Pw=	0.00	
20.	1D-C2_focus_lamr2_mu3	K=	-2.47	avgD=	4.04	Fin=-2.25	Fb=	0.00	Pw=	0.00	

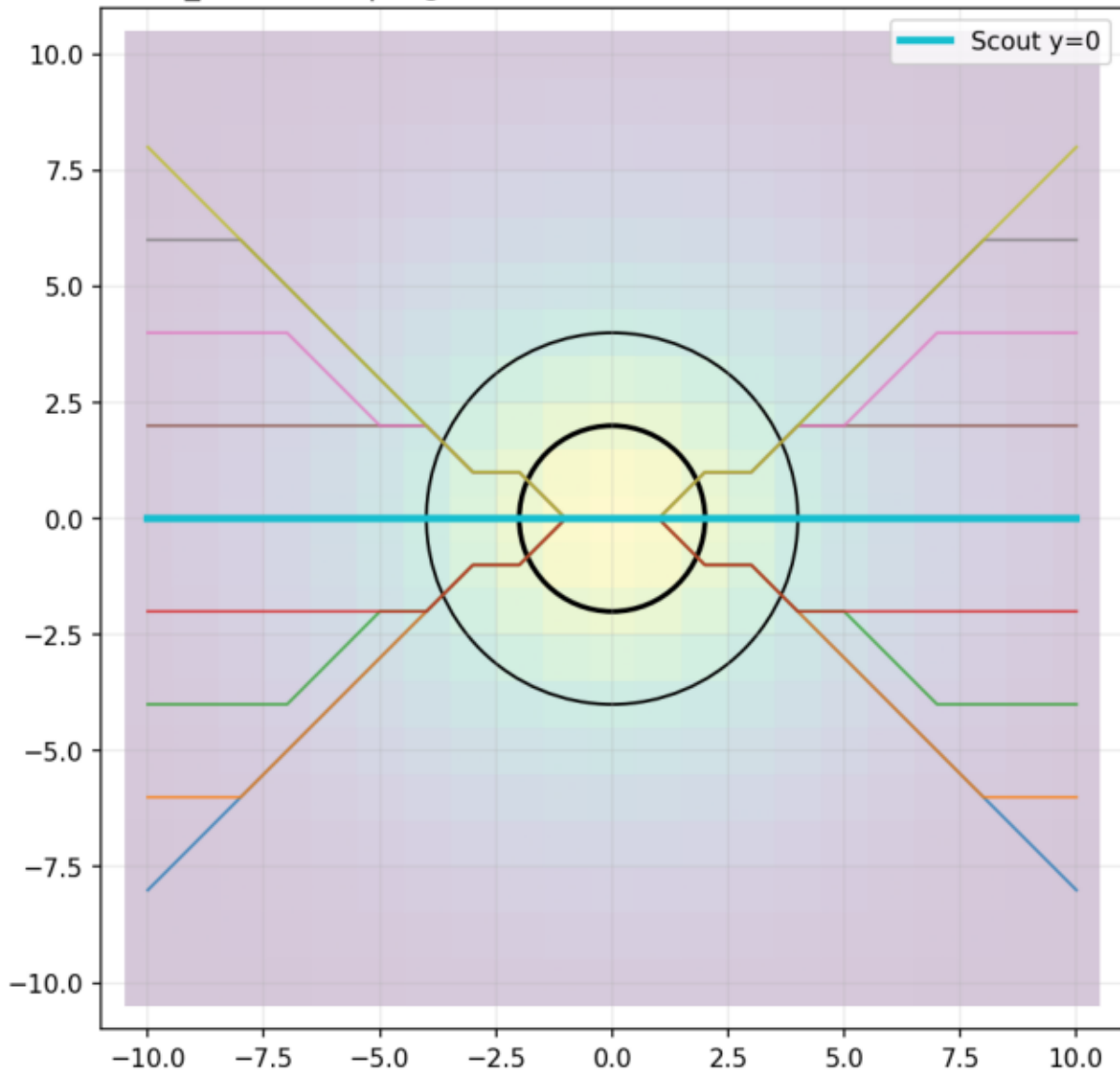
1D-A_lambda10
K_v1=-2.75 | avgD=3.58, Fin=-2.75, Fb=0.00, Pw=0.00



1D-C1_l1_m3
K_v1=0.14 | avgD=2.18, Fin=1.25, Fb=0.00, Pw=0.00



1D-C3_connect_mu2
K_v1=-35.50 | avgD=-5.20, Fin=5.00, Fb=16.00, Pw=25.99



1D-C3_lamr2_mu3
K_v1=-18.16 | avgD=-2.62, Fin=3.00, Fb=8.00, Pw=13.10

